ShopEase -E-commerce Platform

A Course Project Report

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INRODUCTION

With the rapid growth of internet technology, e-commerce platforms have become an integral part of modern shopping. They allow users to browse products, compare options, and make purchases conveniently from anywhere. This has increased the demand for well-designed, user-friendly, and scalable online shopping systems.

ShopEase – E-commerce Platform is a UI/UX design project created using Figma, aimed at developing a modern and responsive e-commerce website. Inspired by platforms like Amazon and Flipkart, the design focuses on providing a smooth and intuitive user experience.

The project covers the complete user journey, including user authentication, product browsing, detailed product views, cart management, and checkout process. It also includes features like search functionality, filtering options, and user account management to enhance usability.

Special emphasis is given to responsive design, consistent visual elements, and reusable components, ensuring the platform works efficiently across desktop, tablet, and mobile devices.

Overall, this project demonstrates a structured approach to designing an end-to-end e-commerce system using modern UI/UX principles and design tools.

PROBLEM STATEMENT

Traditional shopping methods require customers to physically visit stores, which is time-consuming and often inconvenient. Shoppers may face limitations such as restricted store timings, limited product availability, difficulty comparing multiple products, and long billing queues. These challenges reduce flexibility and efficiency, especially for users who prefer quick and convenient purchasing options.

Although e-commerce platforms have transformed retail by enabling online shopping, many existing systems still suffer from usability issues. Complex navigation structures, inconsistent interface designs, cluttered layouts, and complicated checkout processes can confuse users and negatively impact their shopping experience. Such problems often result in increased cart abandonment rates and reduced customer satisfaction.

Modern users expect seamless digital experiences that allow them to easily search for products, apply filters, view detailed product information, and complete transactions securely. Additionally, customers access online shopping platforms from various devices such as smartphones, tablets, and desktops. Lack of responsive design and consistency across devices further creates accessibility and usability challenges.

Therefore, there is a need to design a modern, user-friendly, and responsive e-commerce platform that simplifies the entire shopping process. The objective of this project is to develop an intuitive UI/UX design that enhances navigation, improves usability, supports secure payment workflows, and ensures a consistent and engaging shopping experience across different devices.

METHODOLOGY

The design of the **ShopEase E-commerce Platform** follows a structured UI/UX design process to ensure a smooth and user-friendly experience.

1. Requirement Analysis

The initial phase of the project involved understanding the problem statement and identifying the core requirements of an e-commerce platform. During this stage, the functional requirements such as user authentication, product browsing, cart management, checkout process, and user dashboard were clearly defined. Additionally, non-functional requirements like responsiveness, accessibility, and performance were also considered to ensure a high-quality user experience.

2. Sitemap Design

A sitemap was created to define the overall structure of the website. It outlines the hierarchy of pages such as Home, Product Catalogue, Cart, Checkout, and User Dashboard, ensuring clear navigation flow.

3. User Flow Design

Following the sitemap, detailed user flows were created to visualize how users interact with the system to complete specific tasks. Key flows such as user login, product search, adding items to the cart, and completing the checkout process were mapped. These flows helped in identifying the sequence of screens and interactions required for a smooth user journey, minimizing confusion and improving usability.

4. Low-Fidelity Wireframing

Low-fidelity wireframes were then developed to establish the basic layout and structure of each screen. These wireframes focused on content placement, navigation elements, and functional components without emphasizing visual design.

5. Design System Development

To maintain consistency across the application, a design system was created. This included defining color schemes, typography, spacing, and reusable UI components.Standard components such as buttons, cards, input fields, navigation bars, and product tiles were designed and stored as reusable elements in Figma. This ensured uniformity and scalability throughout the project.

6. High-Fidelity UI Design

Once the design system was established, high-fidelity UI screens were created to represent the final look and feel of the application. These screens included detailed visual elements, images, icons, and real content.

7. Prototyping and Interaction Design

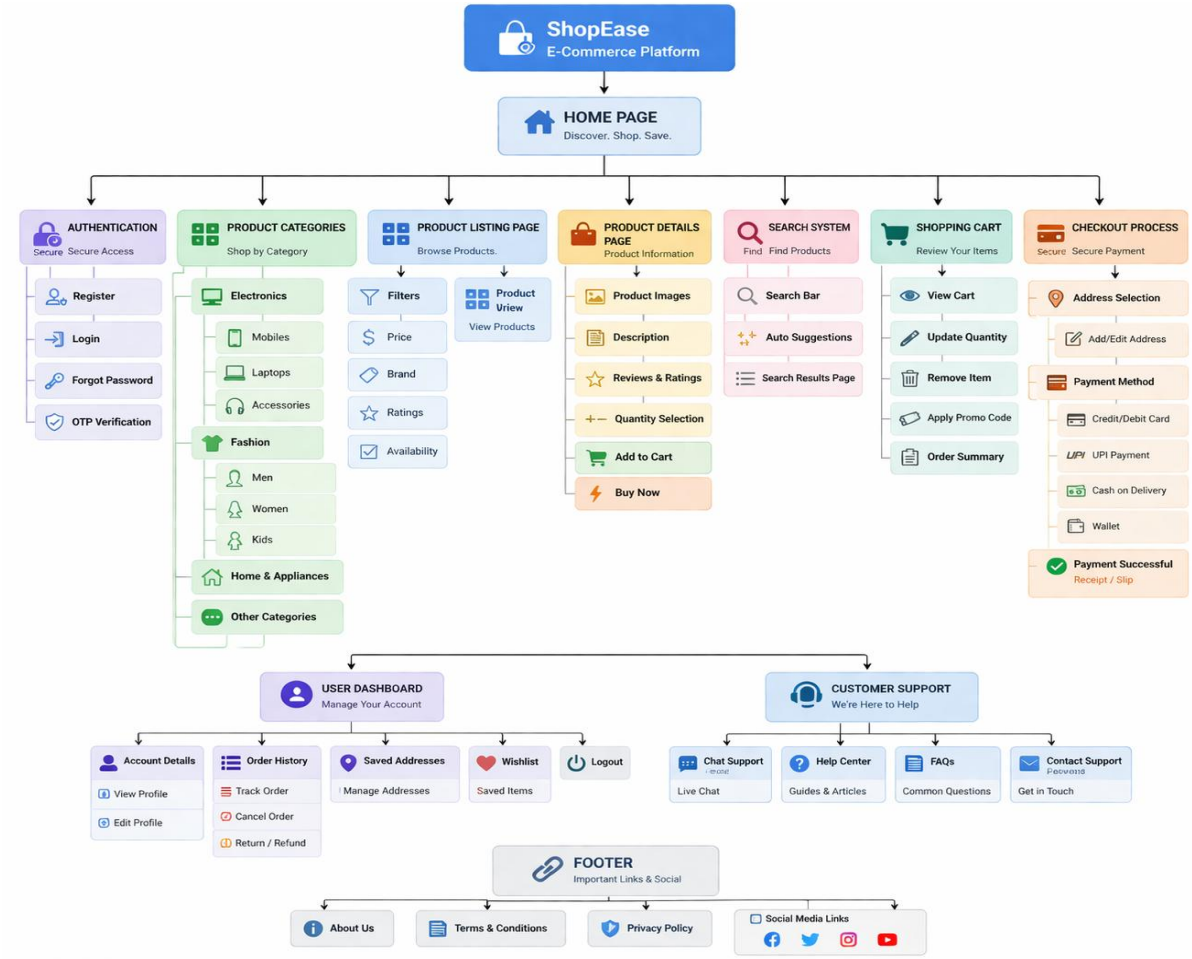
Interactive prototypes were developed in Figma to simulate real user interactions. Navigation links were added between screens to demonstrate user flows such as adding a product to the cart or completing a purchase. Micro-interactions like button hover effects, loading states, and success messages were also incorporated to enhance user engagement.

8. Responsive Design Implementation

The design was made responsive to ensure compatibility across different devices, including desktops, tablets, and mobile phones. Techniques such as flexible grids, auto layout, and adaptive components were used to adjust content dynamically. This ensured a seamless user experience regardless of the device being used.

9. Testing and Validation

The final phase involved testing the design for usability, accessibility, and consistency. The navigation flow was reviewed to ensure that users could complete tasks without difficulty. Accessibility factors such as color contrast, readable font sizes, and clear visual hierarchy were verified. Feedback was incorporated to refine the design and eliminate any usability issues.



User Flow Diagram

IMPLEMENTATION

The **ShopEase E-commerce Platform** is designed using Figma as a UI/UX design and prototyping tool. This project focuses on the visual design and user interaction flow rather than backend development or actual coding implementation.

Design Implementation

The following features were implemented within Figma:

- **Auto Layout:** Used to create responsive and flexible designs that adapt to different screen sizes.
- **Component-Based Design:** Reusable components such as buttons, cards, forms, and navigation bars were created to maintain consistency.
- **Prototyping:** Interactive links were added between screens to simulate real-time user navigation and workflow.
- **Design System:** A consistent set of colors, typography, and UI elements was defined and applied throughout the project.

Functional Implementation (UI Level)

Although no backend code is written, the following functionalities are visually represented:

- User authentication screens (Login, Register, OTP verification)
- Product browsing and filtering system
- Product detail view with interactive elements
- Cart management and checkout flow
- Payment process and success confirmation
- User dashboard and order tracking

CODE

Home.tsx

```
import { useState, useEffect } from 'react';
import { Link } from 'react-router';
import { motion } from 'motion/react';
import { ChevronLeft, ChevronRight, ArrowRight } from 'lucide-react';
import ProductCard from '../components/ProductCard';
import Button from '../components/Button';
const banners = [
 {
   id: 1,
   title: 'Summer Sale',
   subtitle: 'Up to 70% off on fashion',
   image: 'https://images.unsplash.com/photo-1641839629833-4838a270e2ce?w=1080',
   link: '/products/fashion',
  },
  {
   id: 2,
   title: 'Latest Electronics',
   subtitle: 'New arrivals at best prices',
   image: 'https://images.unsplash.com/photo-1760587162690-95608c8ab2da?w=1080',
   link: '/products/electronics',
  },
  {
   id: 3,
   title: 'Home Makeover',
   subtitle: 'Transform your space',
   image: 'https://images.unsplash.com/photo-1662059361834-d361807d63e7?w=1080',
   link: '/products/home-kitchen',
  },
];
const categories = [
  { name: 'Electronics', image: 'https://images.unsplash.com/photo-1758186374131-d542d2beae0c?w=400',
link: '/products/electronics' },
  { name: 'Fashion', image: 'https://images.unsplash.com/photo-1641839588934-6d481144605b?w=400', link:
'/products/fashion' },
  { name: 'Home & Kitchen', image: 'https://images.unsplash.com/photo-1567016526105-
22da7c13161a?w=400', link: '/products/home-kitchen' },
  { name: 'Beauty', image: 'https://images.unsplash.com/photo-1660324579818-0160263f0d55?w=400', link:
'/products/beauty' },
```

```
  { name: 'Sports', image: 'https://images.unsplash.com/photo-1636614223954-db6a663293ef?w=400', link:
'/products/sports' },
  { name: 'Books', image: 'https://images.unsplash.com/photo-1633557614023-f37613f7c4b7?w=400', link:
'/products/books' },
];
const trendingProducts = [
  {
    id: '1',
    name: 'iPhone 15 Pro Max',
    price: 134900,
    originalPrice: 159900,
    discount: 15,
    rating: 4.5,
    reviews: 2453,
    image: 'https://images.unsplash.com/photo-1758186374131-d542d2beae0c?w=400',
  },
  {
    id: '2',
    name: 'MacBook Pro 14"',
    price: 189900,
    originalPrice: 209900,
    discount: 10,
    rating: 4.8,
    reviews: 1876,
    image: 'https://images.unsplash.com/photo-1636614223954-db6a663293ef?w=400',
  },
  {
    id: '3',
    name: 'Sony WH-1000XM5',
    price: 29990,
    originalPrice: 34990,
    discount: 14,
    rating: 4.7,
    reviews: 3201,
    image: 'https://images.unsplash.com/photo-1630148198235-6bd561ba72a4?w=400',
  },
  {
    id: '4',
    name: 'Premium Leather Jacket',
    price: 8999,
```

```
  originalPrice: 14999,
  discount: 40,
  rating: 4.3,
  reviews: 567,
  image: 'https://images.unsplash.com/photo-1629373915883-944ae752da19?w=400',
},
{
  id: '5',
  name: 'Designer Watch',
  price: 12499,
  originalPrice: 19999,
  discount: 37,
  rating: 4.6,
  reviews: 892,
  image: 'https://images.unsplash.com/photo-1760520338238-4137dd2dc28f?w=400',
},
{
  id: '6',
  name: 'Modern Armchair',
  price: 15999,
  originalPrice: 24999,
  discount: 36,
  rating: 4.4,
  reviews: 234,
  image: 'https://images.unsplash.com/photo-1567016526105-22da7c13161a?w=400',
},
{
  id: '7',
  name: 'Smart Watch Pro',
  price: 24999,
  originalPrice: 29999,
  discount: 17,
  rating: 4.5,
  reviews: 1543,
  image: 'https://images.unsplash.com/photo-1760520338238-4137dd2dc28f?w=400',
},
{
  id: '8',
  name: 'Wireless Earbuds',
  price: 7999,
```

```
    originalPrice: 12999,
    discount: 38,
    rating: 4.2,
    reviews: 2876,
    image: 'https://images.unsplash.com/photo-1761641466573-f240b6e446de?w=400',
  },
];
export default function Home() {
  const [currentBanner, setCurrentBanner] = useState(0);
  useEffect(() => {
    const timer = setInterval(() => {
      setCurrentBanner((prev) => (prev + 1) % banners.length);
    }, 5000);
    return () => clearInterval(timer);
  }, []);
  const nextBanner = () => {
    setCurrentBanner((prev) => (prev + 1) % banners.length);
  };
  const prevBanner = () => {
    setCurrentBanner((prev) => (prev - 1 + banners.length) % banners.length);
  };
return (
    <div>
      <section className="relative h-[500px] overflow-hidden bg-muted">
        {banners.map((banner, index) => (
          <motion.div
            key={banner.id}
            className="absolute inset-0"
            initial={{ opacity: 0 }}
            animate={{
              opacity: currentBanner === index ? 1 : 0,
              scale: currentBanner === index ? 1 : 1.1,
            }}
            transition={{ duration: 0.7 }}
          >
            <img
              src={banner.image}
              alt={banner.title}
              className="h-full w-full object-cover"
            />          <div className="absolute inset-0 bg-gradient-to-r from-black/60 to-transparent">
```

```
          <div className="mx-auto max-w-7xl px-4 h-full flex items-center">
            <motion.div
              className="max-w-xl text-white"
              initial={{ opacity: 0, x: -50 }}
              animate={{
                opacity: currentBanner === index ? 1 : 0,
                x: currentBanner === index ? 0 : -50,
              }}
              transition={{ duration: 0.7, delay: 0.2 }}
            >
              <h1 className="mb-4 text-5xl text-white">{banner.title}</h1>
              <p className="mb-8 text-xl text-white/90">{banner.subtitle}</p>
              <Link to={banner.link}>
                <Button size="lg">
                  Shop Now <ArrowRight className="h-5 w-5" />
                </Button>
              </Link>
            </motion.div>
          </div>
        </div>
      </motion.div>
    ))}
    <button
      onClick={prevBanner}
      className="absolute left-4 top-1/2 -translate-y-1/2 flex h-12 w-12 items-center justify-center rounded-full bg-white/20 text-white backdrop-blur-sm hover:bg-white/30 transition-colors" >
      <ChevronLeft className="h-6 w-6" />
    </button>
    <button
      onClick={nextBanner}
      className="absolute right-4 top-1/2 -translate-y-1/2 flex h-12 w-12 items-center justify-center rounded-full bg-white/20 text-white backdrop-blur-sm hover:bg-white/30 transition-colors"
    >
      <ChevronRight className="h-6 w-6" />
    </button>
    <div className="absolute bottom-4 left-1/2 -translate-x-1/2 flex gap-2">
      {banners.map((_, index) => (
        <button
          key={index}
          onClick={() => setCurrentBanner(index)}
```

```
            className={`h-2 rounded-full transition-all ${
              currentBanner === index ? 'w-8 bg-white' : 'w-2 bg-white/50'
            }`}
          />
        ))}
      </div>
    </section>

    <section className="mx-auto max-w-7xl px-4 py-12">
      <h2 className="mb-6">Shop by Category</h2>
      <div className="grid grid-cols-2 md:grid-cols-3 lg:grid-cols-6 gap-4">
        {categories.map((category) => (
          <Link
            key={category.name}
            to={category.link}
            className="group block overflow-hidden rounded-lg border border-border bg-card hover:shadow-lg
transition-shadow"
          >
            <div className="aspect-square overflow-hidden">
              <img
                src={category.image}
                alt={category.name}
                className="h-full w-full object-cover transition-transform group-hover:scale-110"
              />
            </div>
            <div className="p-3 text-center">
              <h3 className="text-sm">{category.name}</h3>
            </div>
          </Link>
        ))}
      </div>
    </section>

    <section className="bg-warning/10 py-12">
      <div className="mx-auto max-w-7xl px-4">
        <div className="mb-6 flex items-center justify-between">
          <div>
            <h2>Deals of the Day</h2>
            <p className="text-muted-foreground">Limited time offers</p>
          </div>
```

```
      <Link to="/products/deals" className="text-primary hover:underline flex items-center gap-1">
        View All <ArrowRight className="h-4 w-4" />
      </Link>
    </div>
    <div className="grid grid-cols-2 md:grid-cols-3 lg:grid-cols-4 gap-4">
      {trendingProducts.slice(0, 4).map((product) => (
        <ProductCard key={product.id} {...product} />
      ))}
    </div>
  </div>
</section>

<section className="mx-auto max-w-7xl px-4 py-12">
  <div className="mb-6 flex items-center justify-between">
    <div>
      <h2>Trending Products</h2>
      <p className="text-muted-foreground">Most popular this week</p>
    </div>
    <Link to="/products/trending" className="text-primary hover:underline flex items-center gap-1">
      View All <ArrowRight className="h-4 w-4" />
    </Link>
  </div>
  <div className="grid grid-cols-2 md:grid-cols-3 lg:grid-cols-4 gap-4">
    {trendingProducts.map((product) => (
      <ProductCard key={product.id} {...product} />
    ))}
  </div>
</section>

<section className="bg-gradient-to-r from-primary to-primary/80 text-primary-foreground py-16">
  <div className="mx-auto max-w-4xl px-4 text-center">
    <h2 className="mb-4 text-white">Join ShopEase Premium</h2>
    <p className="mb-8 text-xl text-white/90">
      Get free shipping, exclusive deals, and early access to sales
    </p>
    <Button variant="secondary" size="lg">
      Learn More
    </Button>
  </div>
</section>
```

```
    </div>  );
}
```

Login.tsx

```
import { useState } from 'react';
import { Link, useNavigate } from 'react-router';
import Button from '../components/Button';
import Input from '../components/Input';
import { Mail, Lock, Eye, EyeOff } from 'lucide-react';
export default function Login() {
  const [showPassword, setShowPassword] = useState(false);
  const [email, setEmail] = useState('');
  const [password, setPassword] = useState('');
  const navigate = useNavigate();
  const handleSubmit = (e: React.FormEvent) => {
    e.preventDefault();
    console.log('Login:', { email, password });
    navigate('/');
  };
  return (
    <div className="min-h-[calc(100vh-200px)] flex items-center justify-center px-4 py-12">
      <div className="w-full max-w-md">
        <div className="text-center mb-8">
          <h1 className="mb-2">Welcome Back</h1>
          <p className="text-muted-foreground">Sign in to your ShopEase account</p>
        </div>
        <div className="rounded-lg border border-border bg-card p-8">
          <form onSubmit={handleSubmit} className="space-y-6">
            <div className="relative">
              <Mail className="absolute left-3 top-[42px] h-5 w-5 text-muted-foreground" />
              <Input
                type="email"
                label="Email Address"
                placeholder="you@example.com"
                value={email}
                onChange={(e) => setEmail(e.target.value)}
                required
                className="pl-10"
              />
```

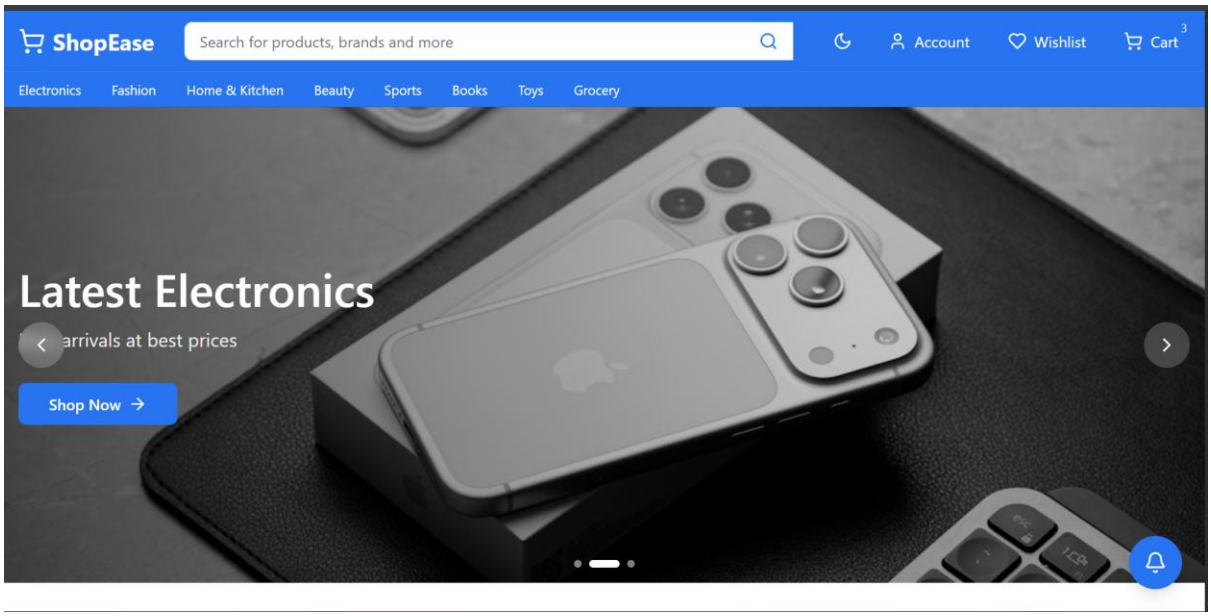
```
  </div>
  <div className="relative">
    <Lock className="absolute left-3 top-[42px] h-5 w-5 text-muted-foreground" />
    <Input
      type={showPassword ? 'text' : 'password'}
      label="Password"
      placeholder="Enter your password"
      value={password}
      onChange={(e) => setPassword(e.target.value)}
      required
      className="pl-10 pr-10"
    />
    <button
      type="button"
      onClick={() => setShowPassword(!showPassword)}
      className="absolute right-3 top-[42px] text-muted-foreground hover:text-foreground"
    >
      {showPassword ? <EyeOff className="h-5 w-5" /> : <Eye className="h-5 w-5" />}
    </button>
  </div>
  <div className="flex items-center justify-between">
    <label className="flex items-center gap-2 text-sm">
      <input type="checkbox" className="rounded border-border" />
      <span>Remember me</span>
    </label>
    <Link
      to="/forgot-password” className="text-sm text-primary hover:underline",  Forgot Password?
    </Link>
  </div>
  <Button type="submit" fullWidth>
    Sign In
  </Button>
</form>
<div className="mt-6">
  <div className="relative">
    <div className="absolute inset-0 flex items-center">
      <div className="w-full border-t border-border"></div>
    </div>
    <div className="relative flex justify-center text-sm">
      <span className="bg-card px-2 text-muted-foreground">Or continue with</span>
```

```
          </div>
        </div>
        <div className="mt-6 grid grid-cols-2 gap-3">
          <Button variant="outline" type="button">
            <svg className="h-5 w-5" viewBox="0 0 24 24">
              <path
                fill="currentColor"
                d="M22.56 12.25c0-.78-.07-1.53-.2-2.25H12v4.26h5.92c-.26 1.37-1.04 2.53-2.21
3.31v2.77h3.57c2.08-1.92 3.28-4.74 3.28 />
              <path
                fill="currentColor"
                d="M12 23c2.97 0 5.46-.98 7.28-2.66l-3.57-2.77c-.98.66-2.23 1.06-3.71 1.06-2.86 0-5.29-1.93-
6.16-4.53H2.18v2.84C3.99 20.53 7.7 23 12 23z"
              />
              <path
                fill="currentColor"
                d="M5.84 14.09c-.22-.66-.35-1.36-.35-2.09s.13-1.43.35-2.09V7.07H2.18C1.43 8.55 1 10.22 1
12s.43 3.45 1.18 4.93l2.85-2.22.81-.62z"
              />
              <path
                fill="currentColor"
                d="M12 5.38c1.62 0 3.06.56 4.21 1.64l3.15-3.15C17.45 2.09 14.97 1 12 1 7.7 1 3.99 3.47 2.18
7.07l3.66 2.84c.87-2.6 3.3-4.53 6.16-4.53z"
              />
            </svg>
            Google
          </Button>         <p className="mt-6 text-center text-sm text-muted-foreground">
        Don't have an account?{' '}
        <Link to="/register" className="text-primary hover:underline">
          Sign up
        </Link>
      </p>
    </div>
  </div>
 </div>
);
}
```

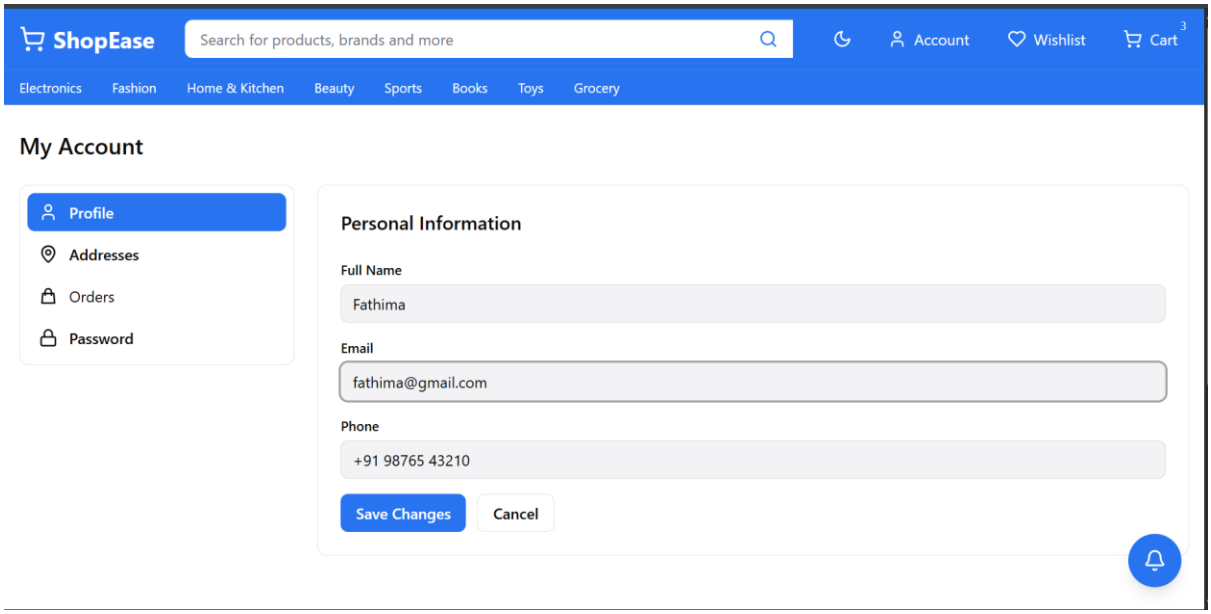
OUTPUT SCREENS

https://group-finite-53614488.figma.site

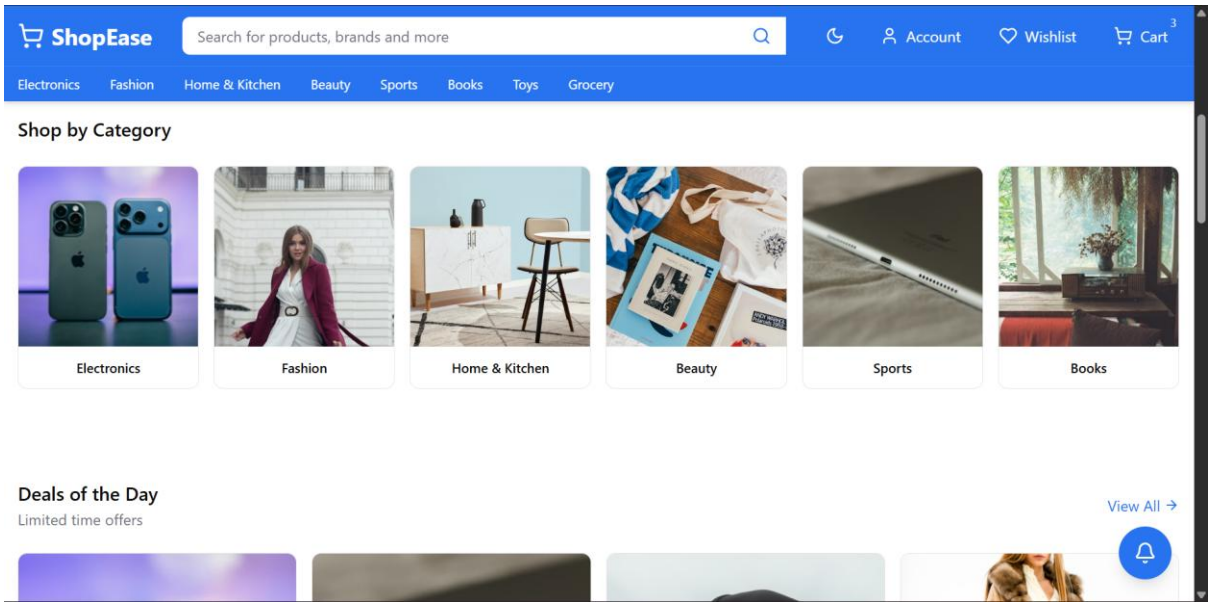
Home Page:



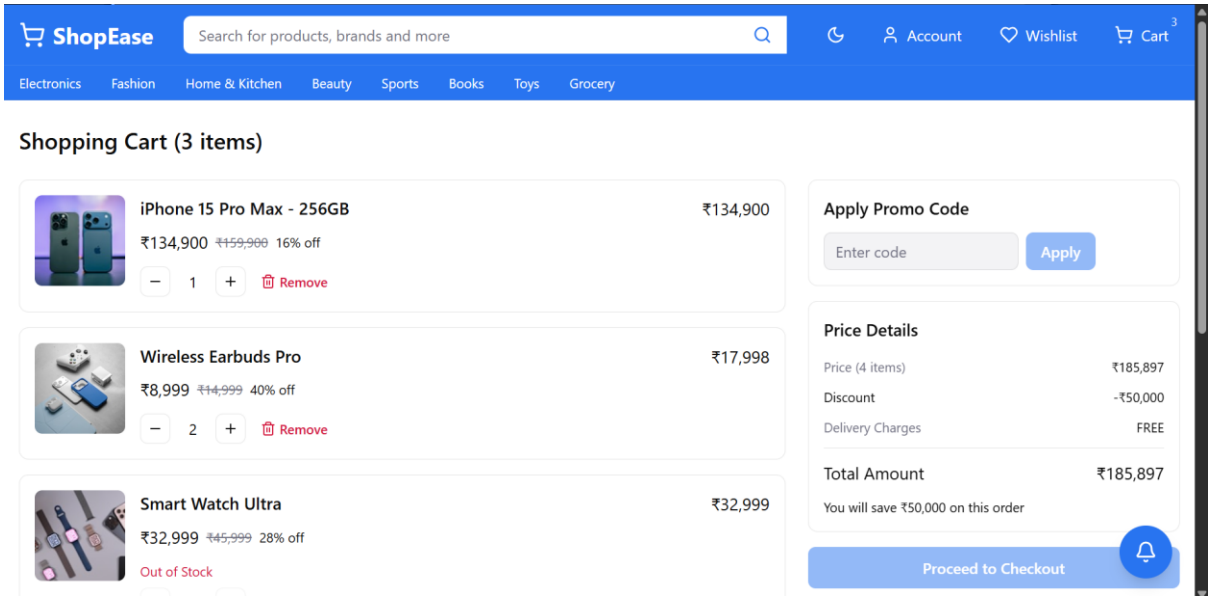
Account Details:



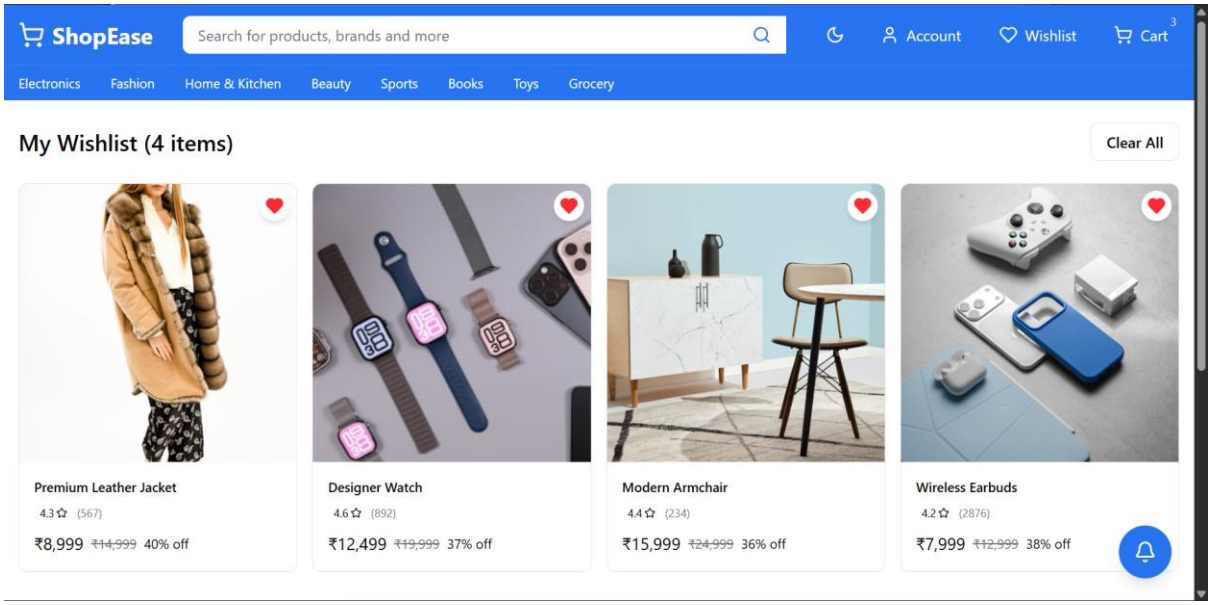
Product Catalogues Page:



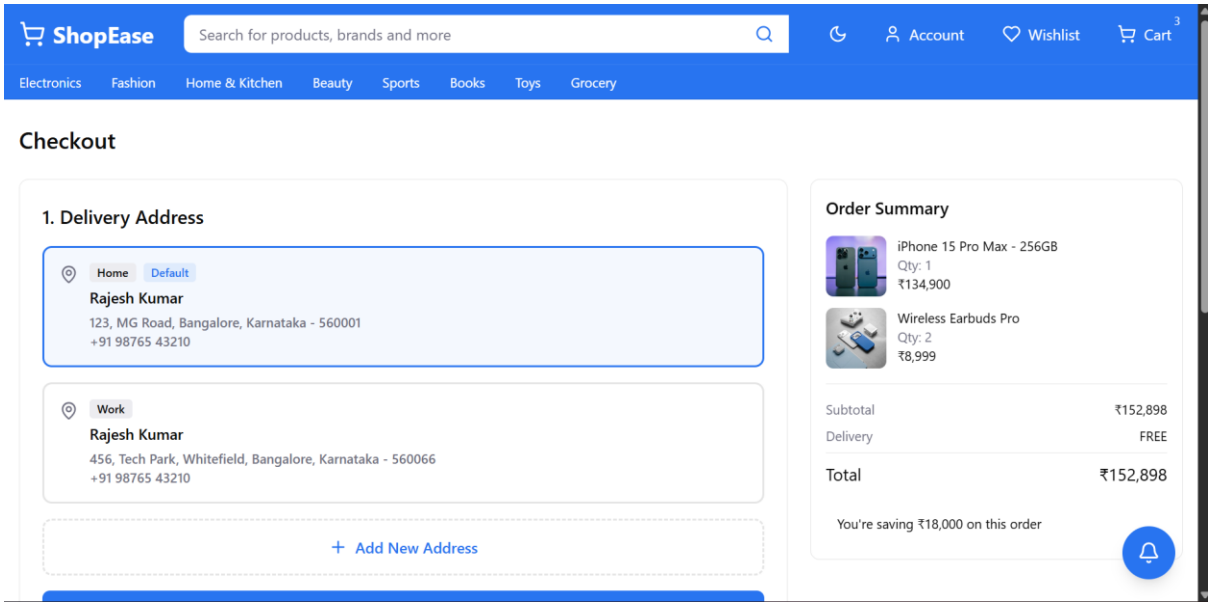
Cart Page:



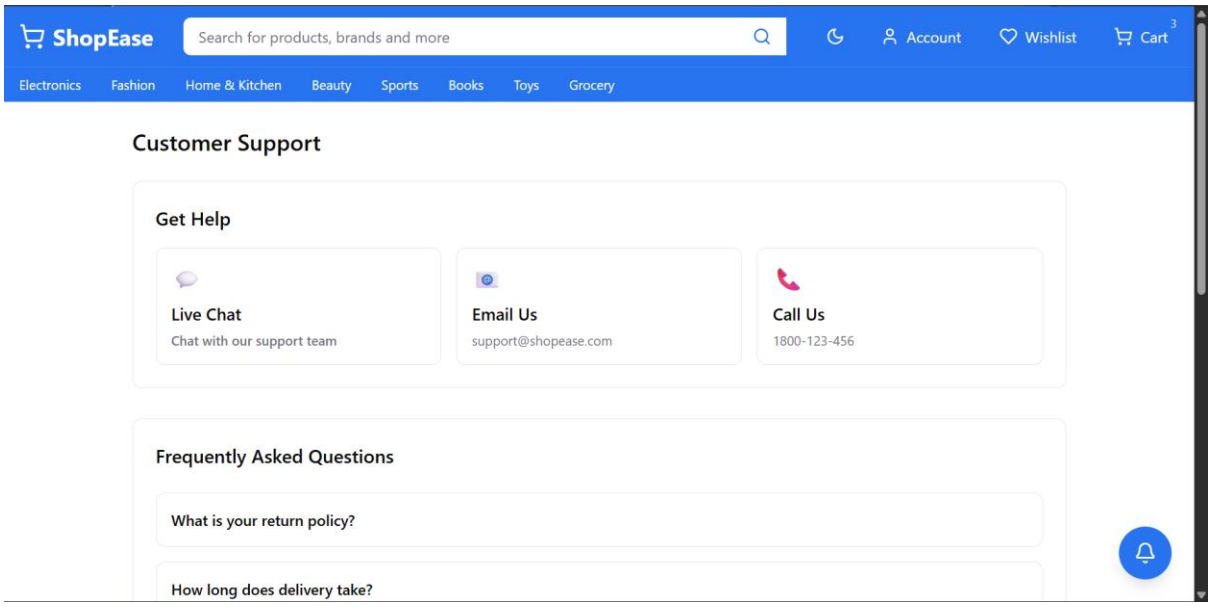
Wishlist Page:



Checkout page:



Search and Support Screens:



REFERENCES

- Figma Official Website – Used for UI/UX design and prototyping
- Amazon Website – Reference for e-commerce features and layout
- Flipkart Website – Reference for UI structure and user experience
- Material Design Guidelines – Used for design principles and UI components
- Bootstrap Documentation – Reference for responsive design concepts